

Lab-IQ-Modulation-and-Demodulation

QPSK like IQ-Modulation and Demodulation

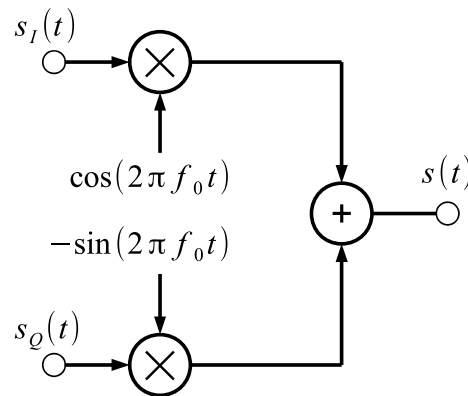
In the next GNUradio flowgraph, IQ-modulation and IQ-demodulation are exemplified by a **QPSK**-like communication system. In **Quadrature Phase-Shift Keying (QPSK)**, the quadrature components, i.e., the inphase component $s_I(t)$ and the quadrature component $s_Q(t)$, which are the real- and the imaginary part of the equivalent lowpass signal $s_L(t)$ of the bandpass transmit signal $s(t)$, are independently generated in a similar manner as the modulated signal $m(t)$ in the bipolar NRZ baseband communication system:

$$s_L(t) = s_I(t) + js_Q(t) = \sum_{k=-\infty}^{\infty} a_I[k]p(t - kT) + j \sum_{k=-\infty}^{\infty} a_Q[k]p(t - kT) = \sum_{k=-\infty}^{\infty} (a_I[k]$$

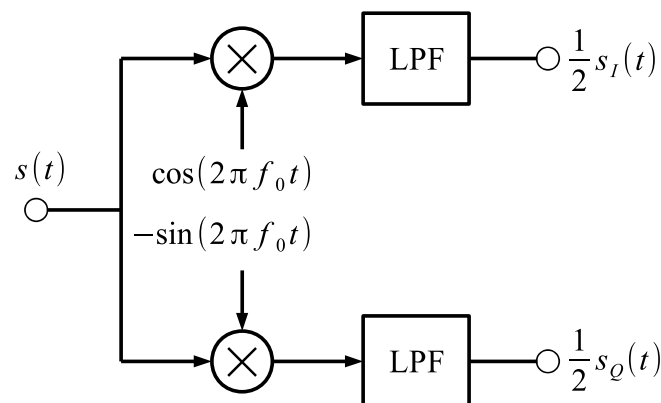
The latter identity holds, since the impulse response $p(t)$ of the root-raised-cosine pulse shaping filter is real-valued. The real-valued bandpass transmit signal $s(t)$ is generated from its equivalent lowpass signal by IQ-modulation:

$$s(t) = \operatorname{Re} \{ s_L(t) \cdot e^{j2\pi f_0 t} \} = s_I(t) \cdot \cos(2\pi f_0 t) - s_Q(t) \cdot \sin(2\pi f_0 t).$$

In the transmitter, this processing, which is also referred to as **upconversion** from baseband to passband, is accomplished by the **IQ modulator**:



The corresponding **downconversion** from passband to baseband can be achieved by the **IQ demodulator**:



where **LPF** corresponds to a suitable **lowpass filter** that **removes the double frequency components around $2f_0$** .

Thus in the receiver, the IQ-components $r_I(t)$ and $r_Q(t)$ of the equivalent baseband signal $r_L(t)$ of the bandpass received signal $r(t)$ are extracted from $r(t)$ by means of IQ-demodulation and multiplication with a **factor of 2**, that corrects the **scaling by $\frac{1}{2}$** introduced by **IQ-modulation and IQ-demodulation**:

$$r_Q(t) = r_I(t) + jr_Q(t) = 2 \cdot \text{LPF} \{r(t) \cdot \cos(2\pi f_0 t)\} - j2 \cdot \text{LPF} \{r(t) \cdot \sin(2\pi f_0 t)\}.$$

Here, $\text{LPF}\{\cdot\}$ again denotes a **lowpass filter** to be applied to the signal in the curly braces, **removing the double frequency components**. The **additional factor of 2** can easily be incorporated into the **filter coefficients** of the

lowpass filter.

If the oscillators that generate the carrier signals $\cos(2\pi f_0 t)$ and $-\sin(2\pi f_0 t)$ in the transmitter as well as in the receiver are phase synchronous, there is no IQ-crosstalk. Then IQ-modulation and -demodulation creates two orthogonal channels, which can be used to transmit two data streams independently. The orthogonality stems from the fact that the two carrier signals $\cos(2\pi f_0 t)$ and $-\sin(2\pi f_0 t)$ are orthogonal. The inphase and the quadrature components $s_I(t)$ and $s_Q(t)$ respectively modulate the amplitude of the cosine and minus sine carrier, which can be used to transmit two data streams without interference between them.

The above equation that relates the bandpass signal $s(t)$ to its equivalent complex lowpass signal $s_L(t)$ in the time domain can be written as:

$$s(t) = \operatorname{Re} \{ s_L(t) \cdot e^{j2\pi f_0 t} \} = [s_L(t)e^{j2\pi f_0 t} + s_L^*(t)e^{-j2\pi f_0 t}] / 2.$$

which using a few properties of the Fourier transform can be used to find the following expression for the corresponding spectra $S(f) \leftrightarrow s(t)$ and $S_L(f) \leftrightarrow s_L(t)$:

$$s(t) = \operatorname{Re} \{ s_L(t)e^{j2\pi f_0 t} \} = [s_L(t)e^{j2\pi f_0 t} + s_L^*(t)e^{-j2\pi f_0 t}] / 2 \quad \Leftrightarrow \quad S(f) = [S_L(f - f_0) +$$

Thus in the frequency domain, the spectrum $S(f)$ of the bandpass signal $s(t)$ can be constructed from the spectrum $S_L(f)$ of the equivalent lowpass signal as follows:

Shift to the right by f_0 and divide by 2 gives positive frequencies part

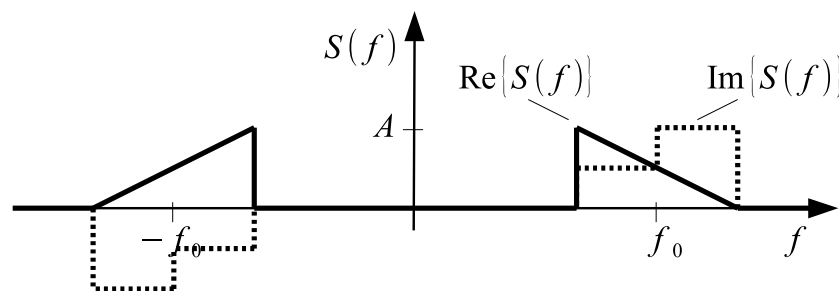
Conjugate symmetry (conjugate & reflect around origin)
then gives negative frequencies part

This means that the spectrum $S_L(f)$ of the equivalent lowpass signal can be derived from the spectrum $S(f)$ of the bandpass signal by applying the corresponding inverse operations in reverse order:

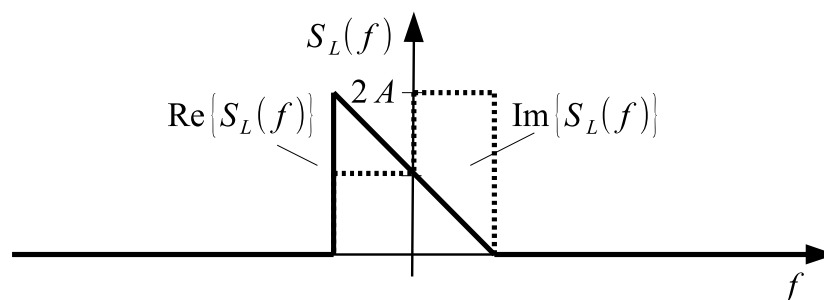
Throw away negative frequencies part

Move positive frequencies part to baseband and scale by 2

This relationship can easily be verified by the following example, where $S(f)$ the spectrum of the real-valued bandpass signal $s(t)$ is depicted in the figure below



and $S_L(f)$ the spectrum of the corresponding complex-valued **baseband signal $s_L(t)$** is given in the next figure:



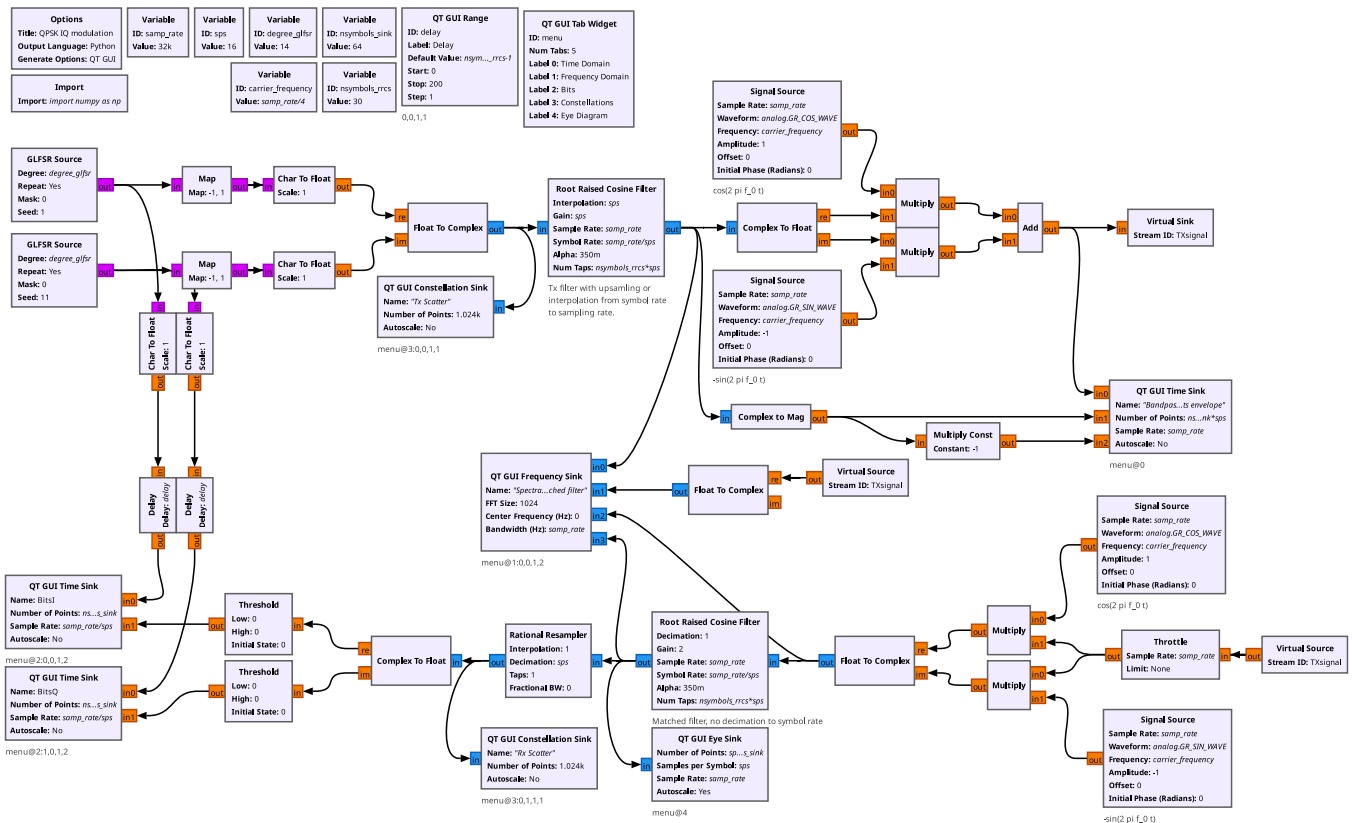
Note that a time-domain signal is real-valued, if its spectrum obeys conjugate symmetry, i.e., the real part is an even function of frequency and the imaginary part is an odd function of frequency. Since $S(f)$ is conjugate symmetric around the origin, the bandpass signal $s(t)$ is

real-valued. Naturally, the bandpass signal always needs to be real-valued because that is the signal that is actually transmitted over the air or over the wire, where complex values are not possible.

However, since $S(f)$ is not conjugate symmetric around the center frequency f_0 , the spectrum of the bandpass signal $S_L(f)$ is not conjugate symmetric and thus the corresponding time domain signal $s_L(t)$ is indeed complex-valued in this case.

GNUradio Flowgraph

To focus on IQ-modulation and demodulation, there is no AWGN channel in the following GNUradio simulation. Open the GNUradio flowgraph QPSK.grc :



Run the GNUradio flowgraph and carefully examine all plotted signals and spectra. Note that different types of

plots are shown in the different tabs of the plot window that opens up.

Answer the following questions and note the answers:

Carefully change the value of the Delay parameter step by step with the arrow key until the stem plots of transmitted and received bits (blue and red color) perfectly align. Then you will only see one stem plot for each symbol sequence of the IQ components.

The spectra shown in the Frequency Domain tab are averaged over multiple FFT outputs. Wait until the spectrum plots have come to an almost steady state before you go on.

What is the name of the samples per symbol parameter N in this flowgraph? What value is it set to?

What is the sampling rate r_s used in this flowgraph?

What is the value of carrier frequency f_0 ?

Which filter acts as the lowpass filter for suppressing the double frequency components?

Select the Frequency Domain tab:

You will find plots of four spectra in one figure window.

Note: You can hide/unhide a plot by clicking of its legend to the right of the actual plot. Thereby, you disable the other plots, if you need to focus on a specific plot!

Hover with the mouse pointer over the maximum of the spectrum $S_L(f)$ of the equivalent lowpass transmit signal $s_L(t)$. (This should be the first curve, plotted in blue color. Disable the other plots.) Read off and note down the shown dB value: $\max |S_l(f)|_{dB} =$

Hover with the mouse pointer over the maximum of the

spectrum $S(f)$ of the transmit signal $s(t)$. (This should be the second curve, plotted in red color. Disable the other plots.) Read off and note down the shown dB value. $\max |S(f)|_{dB} =$

Determine the difference $\max |S_L(f)|_{dB} - |S(f)|_{dB}$ and round this to an integer dB value: $\max |S_L(f)|_{dB} - |S(f)|_{dB} =$

Convert the dB difference to an amplitude relationship: $\max |S_L(f)| / |S(f)| =$

Hover with the mouse pointer over the maximum of the spectrum $R_L(f)$ of the equivalent lowpass receive signal $r_L(t)$. Read off and note down the shown dB value. $\max |R_L(f)|_{dB} =$

Determine the difference $\max |S_L(f)|_{dB} - |R_L(f)|_{dB}$ and round this to an integer dB value:

$\max |S_L(f)|_{dB} - |R_L(f)|_{dB} =$

Convert the dB difference to an amplitude relationship: $\max |S_L(f)| / |R_L(f)| =$

As can be seen from the scatter diagrams of the transmitted and received symbols plotted in the Constellation tab, this amplitude loss must be compensated somewhere in the receiver! Where?

